

Bayesian Regression and Ridge Regression

Goal

Add a **Bayesian prior** to linear regression.

Result: **ridge regression** — more robust when predictors are nearly collinear.

The Linear Model

$$Y = XM + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

OLS gives the **maximum likelihood estimate** of M .

A Prior on the Coefficients

Assume the coefficients are themselves random:

$$M \sim \mathcal{N}(0, \tau^2)$$

Interpretation: *a priori*, we expect the coefficients to be small.

Bayes' Theorem

$$P(M \mid Y, X) = \frac{P(Y, X \mid M) P(M)}{P(Y, X)}$$

We want the **posterior** distribution of M given the data.

The Posterior

$$P(M \mid Y, X) = A \exp\left(-\frac{\|Y - XM\|^2}{2\sigma^2}\right) \exp\left(-\frac{\|M\|^2}{2\tau^2}\right)$$

This is a **normal distribution** in M .

The Posterior, Interpreted

The two factors in the posterior:

▶ $e^{-\|Y - XM\|^2 / 2\sigma^2}$ — **likelihood** (fit to data)

▶ $e^{-\|M\|^2 / 2\tau^2}$ — **prior** (preference for small M)

MAP Estimate: Setting Up

Maximize the posterior $\Leftrightarrow$ set gradient to zero:

$$\frac{X^T Y - (X^T X) M_r}{\sigma^2} - \frac{M_r}{\tau^2} = 0$$

MAP Estimate: The Ridge Formula

Let $s = \sigma^2/\tau^2$. Rearranging gives:

$$(X^T X + sI) M_r = X^T Y$$

The **ridge regression coefficients**:

$$M_r = (X^T X + sI)^{-1} X^T Y$$

Practical Use: Standardize First

Ridge regression is **not scale-invariant**.

Standard practice: center and standardize before fitting.

$$x_i \mapsto \frac{x_i - \mu_i}{\sigma_i}, \quad y \mapsto \frac{y - \mu}{\sigma}$$

Call the standardized data matrix X_0 . Treat $s \geq 0$ as a tuning parameter.

Effect on Invertibility

The covariance matrix $D_0 = \frac{1}{N}X_0^T X_0$ may be singular.

Ridge regression replaces ND_0 by $ND_0 + sI$.

Eigenvalues of $ND_0 + sI$ are $N\lambda_i + s \geq s > 0$.

$\Rightarrow$ **always invertible.**

Condition Number

The **condition number** of a matrix measures numerical sensitivity:

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$$

Matrix	Condition number
ND_0	λ_1/λ_k
$ND_0 + sI$	$(N\lambda_1 + s)/(N\lambda_k + s)$

Larger $s \Rightarrow$ smaller condition number $\Rightarrow$ more stable.

Deriving the SVD: Starting Point

By the **Spectral Theorem**, since D_0 is real symmetric:

$$D_0 = P\Lambda P^T$$

P orthogonal, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$.

Therefore:

$$X_0^T X_0 = N D_0 = P(N\Lambda)P^T$$

Deriving the SVD: Orthogonal Columns

Define $\tilde{\Lambda}_0 = \text{diag}(\sqrt{N\lambda_1}, \dots, \sqrt{N\lambda_k})$, so $X_0^\top X_0 = P\tilde{\Lambda}_0^2 P^\top$.

Look at the matrix $X_0 P$:

$$(X_0 P)^\top (X_0 P) = P^\top (P\tilde{\Lambda}_0^2 P^\top) P = \tilde{\Lambda}_0^2$$

The columns of $X_0 P$ are **orthogonal** with norms $\sqrt{N\lambda_i}$.

Deriving the SVD: Constructing U

Normalize: define $U = X_0 P \tilde{\Lambda}_0^{-1}$.

Check:

$$U^T U = \tilde{\Lambda}_0^{-1} (X_0 P)^T (X_0 P) \tilde{\Lambda}_0^{-1} = \tilde{\Lambda}_0^{-1} \tilde{\Lambda}_0^2 \tilde{\Lambda}_0^{-1} = I_k$$

The columns of U are **orthonormal**.

The SVD of X_0

From $U = X_0 P \tilde{\Lambda}_0^{-1}$:

$$\underset{N \times k}{X_0} = \underset{N \times k}{U} \underset{k \times k}{\tilde{\Lambda}_0} \underset{k \times k}{P^T}$$

- ▶ U — orthonormal columns (left singular vectors)
- ▶ $\tilde{\Lambda}_0$ — diagonal, entries $\sqrt{N\lambda_i}$ (singular values)
- ▶ P — orthogonal (right singular vectors / principal directions)

OLS via the SVD

The columns $u_1, \dots, u_k$ of U form an orthonormal basis for the column space of X_0 .

OLS predicted values:

$$\hat{Y} = \sum_{j=1}^k (u_j \cdot Y) u_j$$

Ridge Prediction via the SVD

Start from $\hat{Y}_r = X_0 M_r$ and substitute the SVD $X_0 = U\tilde{\Lambda}_0 P^\top$:

$$\hat{Y}_r = U\tilde{\Lambda}_0 P^\top (P\tilde{\Lambda}_0^\top U^\top U\tilde{\Lambda}_0 P^\top + sI)^{-1} P\tilde{\Lambda}_0^\top U^\top Y$$

Since $U^\top U = I$ and $P^\top P = I$, and writing $\Lambda = \tilde{\Lambda}_0^2$, this simplifies to:

$$\hat{Y}_r = U\tilde{\Lambda}_0 (\Lambda + sI)^{-1} \tilde{\Lambda}_0 U^\top Y$$

Ridge Prediction: The Middle Factor

All three matrices $\tilde{\Lambda}_0$, $(\Lambda + sI)^{-1}$, $\tilde{\Lambda}_0$ are $k \times k$ diagonal.

Their product is diagonal with j -th entry:

$$\sqrt{N\lambda_j} \cdot \frac{1}{N\lambda_j + s} \cdot \sqrt{N\lambda_j} = \frac{N\lambda_j}{N\lambda_j + s}$$

Define $W = \tilde{\Lambda}_0(\Lambda + sI)^{-1}\tilde{\Lambda}_0 = \text{diag}\left(\frac{N\lambda_j}{N\lambda_j + s}\right)$.

Ridge Prediction: Expanding $UWU^T Y$

So $\hat{Y}_r = UWU^T Y$ where $W = \text{diag}(w_1, \dots, w_k)$.

Writing $U = [u_1 \ \cdots \ u_k]$, we have $UWU^T = \sum_j w_j u_j u_j^T$.

Therefore:

$$\hat{Y}_r = \sum_{j=1}^k w_j u_j u_j^T Y = \sum_{j=1}^k w_j (u_j \cdot Y) u_j$$

Ridge Prediction: Component Form

$$\hat{Y}_r = \sum_{j=1}^k \frac{N\lambda_j}{N\lambda_j + s} (u_j \cdot Y) u_j$$

Compare with OLS:

$$\hat{Y} = \sum_{j=1}^k (u_j \cdot Y) u_j$$

Each term is multiplied by the **shrinkage factor** $\frac{N\lambda_j}{N\lambda_j + s} \leq 1$.

The Shrinkage Factors

$$w_j = \frac{N\lambda_j}{N\lambda_j + s}$$

- ▶ λ_j large $\Rightarrow w_j \approx 1$ (high-variance direction, little shrinkage)
- ▶ λ_j small $\Rightarrow w_j \approx 0$ (low-variance direction, strong shrinkage)
- ▶ $s \rightarrow 0$: ridge $\rightarrow$ OLS
- ▶ $s \rightarrow \infty$: $\hat{Y}_r \rightarrow 0$

Why “Shrinkage”?

Ridge regression **downweights** directions of low variance.

These are precisely the directions where the data give least information — and where OLS is most unstable.

Shrinking them reduces variance at the cost of a small bias.

Summary

Prior	$M \sim \mathcal{N}(0, \tau^2)$
Ridge formula	$M_r = (X^\top X + sI)^{-1} X^\top Y$
Parameter	$s = \sigma^2 / \tau^2$, tuned in practice
Standardize	always, before fitting
Stability	condition number decreases with s
Shrinkage	low-variance components downweighted